

Repeated Integration and Rvachev's Up-Function

Exact finite splines, Thue–Morse structure, and sharp convergence

Complementary report to *The Fabius Function and Rvachev's Up-Function*
Vladimir Reshetnikov's ProveIt repository

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Abstract

A Math Stack Exchange question asks for a closed form for the sequence

$$f_0(x) = \frac{1}{2} \mathbf{1}_{(-1,1)}(x), \quad f_n(x) = 2 \int_{-1}^x (f_{n-1}(2t+1) - f_{n-1}(2t-1)) dt.$$

The first integration produces the piecewise linear tent $f_1(x) = (1 - |x|)_+$, and the subsequent iterates visually converge very rapidly to Rvachev's compactly supported C^∞ up-function. This report gives a complete finite-stage description and a sharp convergence theorem.

The integral operator is exactly a probabilistic averaging operator: if g is the density of Y and U is uniform on $[-1, 1]$, then the next density is that of $(Y + U)/2$. Consequently, for $n \geq 1$,

$$f_n = u_{2^{-1}} * u_{2^{-2}} * \cdots * u_{2^{-n}} * u_{2^{-n}}, \quad u_a(x) = \frac{1}{2a} \mathbf{1}_{[-a,a]}(x).$$

This identifies f_n as a nonuniform box spline. Collapsing its general inclusion–exclusion formula with the Thue–Morse generating polynomial yields, for $n \geq 1$,

$$f_n(x) = \frac{2^{n(n+1)/2-1}}{n!} \sum_{j=0}^{2^n} (\tau_n(j) - \tau_n(j-1)) \left(x + 1 - \frac{j}{2^{n-1}} \right)_+^n,$$

where $\tau_n(j) = (-1)^{\text{wt}(j)}$ on $0 \leq j < 2^n$ and is zero outside that range. The n th derivative is therefore a rescaled Thue–Morse step function.

The principal quantitative result derived here is exact, rather than merely asymptotic. For every fixed $r \geq 0$ and every $n \geq r + 2$,

$$\left\| f_n^{(r)} - \text{up}^{(r)} \right\|_\infty = \frac{2^{\binom{r+3}{2}}}{9 \cdot 4^n}.$$

Thus, once the r th derivative has passed its first admissible spline stage, every further integration divides its uniform error by exactly four. In the indexing that starts with the piecewise linear tent $g_0 = f_1$, this becomes

$$\|g_m - \text{up}\|_\infty = \frac{2}{9 \cdot 4^m} \quad (m \geq 1),$$

while the exceptional initial error is $\|g_0 - \text{up}\|_\infty = 13/72$. The proof is based on an exact Peano-kernel identity. The comparison kernel between a uniform random variable and the up-distributed random variable has both integral and maximum equal to $1/9$; the finite partial density has disjoint extremal plateaux whose height is known exactly. Their combination gives both the upper bound and matching extremizers.

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1 The problem, the indexing, and the answer

The recurrence considered in the Math Stack Exchange question [1, 2] is

$$f_0(x) = \frac{1}{2} \mathbf{1}_{(-1,1)}(x), \quad f_n(x) = 2 \int_{-1}^x (f_{n-1}(2t+1) - f_{n-1}(2t-1)) dt. \quad (1)$$

Changing the values of f_0 at $x = \pm 1$ changes neither any later iterate nor any of the norm statements below. We therefore use whichever endpoint convention is convenient.

There is a small indexing issue in the informal description of the construction. The literal seed f_0 in (1) is piecewise constant; the first iterate is the piecewise linear function

$$f_1(x) = (1 - |x|)_+. \quad (2)$$

When one says that the up-function arises by “repeatedly integrating a piecewise linear function,” the natural indexing is therefore

$$g_m = f_{m+1}, \quad g_0(x) = (1 - |x|)_+. \quad (3)$$

Both indexings will be retained: f_n matches the question, while g_m matches the piecewise-linear wording.

1.1 Main finite-stage formula

Write $\text{wt}(j)$ for the number of 1's in the binary expansion of j . For $n \geq 0$ define the zero-extended finite Thue–Morse word

$$\tau_n(j) = \begin{cases} (-1)^{\text{wt}(j)}, & 0 \leq j < 2^n, \\ 0, & \text{otherwise,} \end{cases} \quad \Delta\tau_n(j) = \tau_n(j) - \tau_n(j-1). \quad (4)$$

The accepted Stack Exchange answer gives a formula equivalent to the following one; the probabilistic and spline derivation in [section 4](#) explains why its coefficients are first differences of the Thue–Morse word.

Theorem 1.1 (Exact n th iterate). *For every $n \geq 1$ and $x \in \mathbb{R}$,*

$$f_n(x) = \frac{2^{n(n+1)/2-1}}{n!} \sum_{j=0}^{2^n} \Delta\tau_n(j) \left(x + 1 - \frac{j}{2^{n-1}} \right)_+^n. \quad (5)$$

Equivalently,

$$f_n(x) = \frac{1}{2n! 2^{n(n-1)/2}} \sum_{j=0}^{2^n} \Delta\tau_n(j) (2^n x + 2^n - 2j)_+^n. \quad (6)$$

The knots are

$$-1, -1 + 2^{1-n}, \dots, 1 - 2^{1-n}, 1. \quad (7)$$

On every open knot interval f_n is a polynomial of degree n , and globally $f_n \in C^{n-1}(\mathbb{R})$.

In the tent indexing (3), [theorem 1.1](#) reads

$$g_m(x) = \frac{2^{m(m+3)/2}}{(m+1)!} \sum_{j=0}^{2^{m+1}} \Delta\tau_{m+1}(j) \left(x + 1 - \frac{j}{2^m} \right)_+^{m+1}. \quad (8)$$

Thus the m th repeated integration of the tent is a degree- $(m+1)$ spline on the dyadic mesh of width 2^{-m} .

1.2 Main sharp convergence theorem

For a bounded function h let $\|h\|_\infty = \sup_{x \in \mathbb{R}} |h(x)|$. When $h \in C^r$, we use the max version of the C^r norm,

$$\|h\|_{C^r} = \max_{0 \leq j \leq r} \|h^{(j)}\|_\infty. \quad (9)$$

Theorem 1.2 (Exact derivative error). *Let $r \geq 0$. The first iterate belonging to C^r is f_{r+1} , and its r th-derivative error is*

$$\|f_{r+1}^{(r)} - \text{up}^{(r)}\|_\infty = \frac{13}{72} 2^{\binom{r+1}{2}}. \quad (10)$$

For every $n \geq r + 2$,

$$\boxed{\|f_n^{(r)} - \text{up}^{(r)}\|_\infty = \frac{2^{\binom{r+3}{2}}}{9 \cdot 4^n}.} \quad (11)$$

All of these bounds are attained at the same 2^{r+2} points

$$c_{r+2,j} = -1 + \frac{2j+1}{2^{r+2}}, \quad 0 \leq j < 2^{r+2}. \quad (12)$$

At the first admissible stage,

$$f_{r+1}^{(r)}(c_{r+2,j}) - \text{up}^{(r)}(c_{r+2,j}) = (-1)^{\text{wt}(j)} \frac{13}{72} 2^{\binom{r+1}{2}}, \quad (13)$$

and for $n \geq r + 2$,

$$f_n^{(r)}(c_{r+2,j}) - \text{up}^{(r)}(c_{r+2,j}) = (-1)^{\text{wt}(j)} \frac{2^{\binom{r+3}{2}}}{9 \cdot 4^n}. \quad (14)$$

The exact quartering begins immediately after the exceptional first C^r spline:

$$\frac{\|f_{r+2}^{(r)} - \text{up}^{(r)}\|_\infty}{\|f_{r+1}^{(r)} - \text{up}^{(r)}\|_\infty} = \frac{4}{13}, \quad \frac{\|f_{n+1}^{(r)} - \text{up}^{(r)}\|_\infty}{\|f_n^{(r)} - \text{up}^{(r)}\|_\infty} = \frac{1}{4} \quad (n \geq r + 2). \quad (15)$$

For the tent indexing this yields the particularly simple uniform result

$$\|g_0 - \text{up}\|_\infty = \frac{13}{72}, \quad \boxed{\|g_m - \text{up}\|_\infty = \frac{2}{9 \cdot 4^m} \quad (m \geq 1).} \quad (16)$$

Thus every integration after the first one gains exactly two binary digits in the sharp uniform error.

The approximation in this report is distinct from the centered finite-CDF spline already discussed in the companion article. Here f_n is itself a probability *density*, and the last unresolved tail is replaced by a scaled uniform variable. The true tail and the uniform replacement have equal mass and equal mean. The first surviving discrepancy is therefore quadratic in the tail scale 2^{-n} , which explains the factor 4^{-n} ; the comparison-kernel argument below determines the coefficient exactly.

2 The integration operator is a probability operator

For $a > 0$, let

$$u_a(x) = \frac{1}{2a} \mathbf{1}_{[-a,a]}(x) \quad (17)$$

be the density of a random variable uniform on $[-a, a]$. Let T denote the operator in (1):

$$(Th)(x) = 2 \int_{-1}^x (h(2t+1) - h(2t-1)) dt. \quad (18)$$

Lemma 2.1 (Moving-window and convolution forms). *Suppose h is integrable and supported in $[-1, 1]$. Then*

$$(Th)(x) = \int_{2x-1}^{2x+1} h(s) \, ds = 2(h * u_1)(2x). \quad (19)$$

Proof. In the first term of (18), put $s = 2t + 1$; in the second, put $s = 2t - 1$. This gives

$$(Th)(x) = \int_{-1}^{2x+1} h(s) \, ds - \int_{-3}^{2x-1} h(s) \, ds.$$

The support assumption allows the lower endpoint -3 in the second integral to be replaced by -1 , after which the difference is the integral over $[2x - 1, 2x + 1]$. The convolution identity follows directly from the definition of u_1 . $\square$

Proposition 2.2 (Probabilistic meaning). *If h is the density of a random variable Y , and U is independent and uniform on $[-1, 1]$, then Th is the density of*

$$\frac{Y + U}{2}. \quad (20)$$

In particular, T preserves nonnegativity, total mass one, evenness, and support in $[-1, 1]$.

Proof. The density of $Y + U$ is $h * u_1$. Scaling a random variable by $1/2$ multiplies its density by 2 and evaluates it at $2x$, giving exactly (19). The listed properties are immediate. $\square$

Let $U_1, U_2, \dots$ be independent random variables, each uniform on $[-1, 1]$. Start with $X_0 = U_1$ and define recursively

$$X_n = \frac{X_{n-1} + U_{n+1}}{2}. \quad (21)$$

Then f_n is the density of X_n .

Theorem 2.3 (Finite random-sum representation). *For every $n \geq 1$,*

$$X_n \stackrel{d}{=} \sum_{k=1}^n 2^{-k} U_k + 2^{-n} U_{n+1}. \quad (22)$$

Consequently,

$$f_n = u_{2^{-1}} * u_{2^{-2}} * \dots * u_{2^{-n}} * u_{2^{-n}}. \quad (23)$$

Proof. Expanding (21) gives weights $2^{-n}, 2^{-n}, 2^{-(n-1)}, \dots, 2^{-1}$ on $n + 1$ independent copies of U . Because the variables are identically distributed, the two copies of 2^{-n} may be placed last, giving (22). Convolution of the corresponding scaled uniform densities gives (23). $\square$

Several facts that are less transparent in the integral recursion are now immediate:

$$f_n \geq 0, \quad \int_{\mathbb{R}} f_n(x) \, dx = 1, \quad f_n(-x) = f_n(x), \quad \text{supp } f_n = [-1, 1]. \quad (24)$$

Moreover, for $n \geq 1$,

$$f_n(0) = \int_{-1}^1 f_{n-1}(s) \, ds = 1. \quad (25)$$

3 The limit is Rvachev's up-function

Consider the almost surely convergent random series

$$X = \sum_{k=1}^{\infty} 2^{-k} U_k. \quad (26)$$

Its support is $[-1, 1]$. Denote its density by up . Existence and smoothness also follow from the Fourier product below; they are classical properties of the Rvachev–Arias de Reyna construction [3].

3.1 Self-similarity and the differential equation

Splitting off the first summand in (26) gives the distributional fixed-point identity

$$X \stackrel{d}{=} \frac{U + X'}{2}, \quad (27)$$

where X' is an independent copy of X . By [theorem 2.2](#), its density is a fixed point of T :

$$\text{up}(x) = \int_{2x-1}^{2x+1} \text{up}(s) \, ds. \quad (28)$$

Differentiating gives

$$\text{up}'(x) = 2(\text{up}(2x+1) - \text{up}(2x-1)). \quad (29)$$

Also $\text{up}(0) = \int_{-1}^1 \text{up} = 1$. The even compactly supported smooth solution of (29) normalized by $\text{up}(0) = 1$ is Rvachev's up-function. Thus the repeated-integration limit is not merely analogous to up ; it is exactly up with the normalization used in the companion article.

3.2 Fourier product

Use the non-unitary characteristic Fourier transform

$$\mathcal{F}h(t) = \int_{\mathbb{R}} h(x) e^{-itx} \, dx, \quad \text{sinc } z = \begin{cases} \sin z/z, & z \neq 0, \\ 1, & z = 0. \end{cases} \quad (30)$$

Then $\mathcal{F}u_a(t) = \text{sinc}(at)$. From (23),

$$\mathcal{F}f_n(t) = \text{sinc}(2^{-n}t) \prod_{k=1}^n \text{sinc}(2^{-k}t) = \left(\prod_{k=1}^{n-1} \text{sinc}(2^{-k}t) \right) \text{sinc}^2(2^{-n}t). \quad (31)$$

Under the symmetric convention used in the Stack Exchange question, the right side is multiplied by $(2\pi)^{-1/2}$, exactly reproducing the conjectured formula there. Taking the limit gives

$$\mathcal{F} \text{up}(t) = \prod_{k=1}^{\infty} \text{sinc}(2^{-k}t). \quad (32)$$

The product is rapidly decreasing together with all its derivatives; Fourier inversion therefore gives a C^∞ density. The product also makes the finite-depth correction transparent: f_n has the first n factors of the limiting product, with the last factor repeated once.

3.3 Relation with the bounded Fabius function

Let $\xi_k = (U_k + 1)/2$, so that the ξ_k are independent uniform variables on $[0, 1]$. Then

$$Y = \sum_{k=1}^{\infty} 2^{-k} \xi_k = \frac{X + 1}{2}. \quad (33)$$

The cumulative distribution function of Y is the bounded Fabius function F . Hence

$$\mathbb{P}(X \leq x) = F\left(\frac{x+1}{2}\right), \quad (34)$$

and, using $F(1 - y) = 1 - F(y)$,

$$\mathbb{P}(X > t) = F\left(\frac{1-t}{2}\right), \quad 0 \leq t \leq 1. \quad (35)$$

On the two halves of the support,

$$\text{up}(x) = \begin{cases} F(x+1), & -1 \leq x \leq 0, \\ F(1-x), & 0 \leq x \leq 1. \end{cases} \quad (36)$$

This is exactly the normalization used in the existing exposition.

4 Exact finite splines and the Thue–Morse collapse

The convolution formula (23) already gives a closed form in the language of box splines. To obtain the explicit truncated-power expression, we first record the general one-dimensional inclusion–exclusion formula.

4.1 A general nonuniform box-spline formula

Lemma 4.1 (Density of a sum of centered uniforms). *Let V_i be independent and uniform on $[-a_i, a_i]$, where $a_i > 0$ and $1 \leq i \leq m$. Put $A = a_1 + \cdots + a_m$. The density b_a of $V_1 + \cdots + V_m$ is*

$$b_a(x) = \frac{1}{2^m(m-1)! \prod_{i=1}^m a_i} \sum_{\varepsilon \in \{0,1\}^m} (-1)^{|\varepsilon|} \left(x + A - 2 \sum_{i=1}^m \varepsilon_i a_i \right)_+^{m-1}. \quad (37)$$

Proof. Translate V_i to $W_i = V_i + a_i$, which is uniform on $[0, 2a_i]$. The distributional derivative of the density of W_i is $(\delta_0 - \delta_{2a_i})/(2a_i)$. After convolving the m factors and integrating $m - 1$ times from the left endpoint, one obtains the usual inclusion–exclusion formula for the density of $W_1 + \cdots + W_m$. Translating the sum back by A gives (37). Equivalently, one may verify the formula by taking Laplace transforms. $\square$

For f_n , the $m = n + 1$ half-widths are

$$2^{-1}, 2^{-2}, \dots, 2^{-(n-1)}, 2^{-n}, 2^{-n}. \quad (38)$$

Their sum is 1, and their product is $2^{-n(n+3)/2}$. Thus the scalar prefactor in (37) becomes

$$\frac{2^{n(n+1)/2-1}}{n!}. \quad (39)$$

It remains to combine subsets that produce the same knot.

4.2 The subset-sum polynomial

After multiplication by 2^{n-1} , the possible increments $2a_i$ in (37) are

$$2^{n-1}, 2^{n-2}, \dots, 2, 1, 1. \quad (40)$$

The signed subset-sum generating polynomial is therefore

$$\begin{aligned} B_n(z) &= (1-z)^2 \prod_{r=1}^{n-1} (1-z^{2^r}) \\ &= (1-z) \prod_{r=0}^{n-1} (1-z^{2^r}). \end{aligned} \quad (41)$$

The classical finite Thue–Morse identity is

$$\prod_{r=0}^{n-1} (1-z^{2^r}) = \sum_{j=0}^{2^n-1} (-1)^{\text{wt}(j)} z^j = \sum_{j \in \mathbb{Z}} \tau_n(j) z^j. \quad (42)$$

Multiplication by $1-z$ takes first differences of the coefficient sequence:

$$B_n(z) = \sum_{j=0}^{2^n} \Delta \tau_n(j) z^j. \quad (43)$$

This explains the otherwise rather mysterious coefficient $c_n(j) - c_n(j-1)$ in the Stack Exchange answer: the duplicated smallest uniform interval contributes the extra factor $1-z$, turning the Thue–Morse word into its first difference.

Proof of theorem 1.1. Insert the widths (38) into theorem 4.1. The subset sum indexed by ε equals $j/2^{n-1}$ after grouping terms with the same integer j ; by (43), the total sign of that group is $\Delta \tau_n(j)$. The scalar is (39). This gives (5). Factoring 2^{-n} from each truncated power gives (6). $\square$

4.3 Derivatives, knots, and exact smoothness

For $0 \leq r \leq n-1$, differentiating (5) gives

$$f_n^{(r)}(x) = \frac{2^{n(n+1)/2-1}}{(n-r)!} \sum_{j=0}^{2^n} \Delta \tau_n(j) \left(x + 1 - \frac{j}{2^{n-1}} \right)_+^{n-r}. \quad (44)$$

The formula is also valid away from the knots for $r = n$. If

$$-1 + \frac{j}{2^{n-1}} < x < -1 + \frac{j+1}{2^{n-1}}, \quad 0 \leq j < 2^n, \quad (45)$$

then the first-difference sum telescopes and gives

$$\boxed{f_n^{(n)}(x) = 2^{n(n+1)/2-1} \tau_n(j).} \quad (46)$$

Thus the highest ordinary derivative is literally a Thue–Morse step word. Because this step function has jumps at the active knots, f_n is C^{n-1} but not C^n globally. Each application of T adds exactly one degree and exactly one order of classical smoothness.

4.4 The first iterates

The first three functions are

$$f_0(x) = \frac{1}{2} \mathbf{1}_{(-1,1)}(x), \quad (47)$$

$$f_1(x) = (1 - |x|)_+, \quad (48)$$

$$f_2(x) = \begin{cases} 1 - 2x^2, & |x| \leq \frac{1}{2}, \\ 2(1 - |x|)^2, & \frac{1}{2} \leq |x| \leq 1, \\ 0, & |x| \geq 1. \end{cases} \quad (49)$$

The next one already displays the full Thue–Morse organization compactly:

$$f_3(x) = \frac{16}{3} \sum_{j=0}^8 \Delta \tau_3(j) \left(x + 1 - \frac{j}{4} \right)_+^3. \quad (50)$$

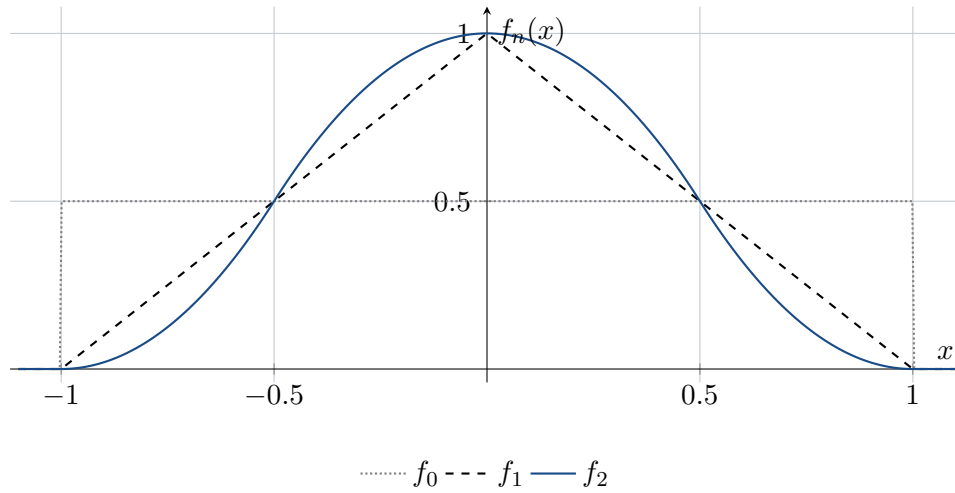


Figure 1: The rectangle, the piecewise linear tent, and the quadratic spline. Later iterates add one polynomial degree and one smoothness order at each step.

5 Replacing the last tail: the exact error decomposition

Put

$$S_n = \sum_{k=1}^n 2^{-k} U_k, \quad a_n = 2^{-n}, \quad (51)$$

and let p_n be the density of S_n . Let X' be an independent copy of the limiting random variable X . The finite and infinite laws split as

$$X_n \stackrel{d}{=} S_n + a_n U, \quad X \stackrel{d}{=} S_n + a_n X'. \quad (52)$$

If

$$\rho_a(x) = a^{-1} \text{up}(x/a) \quad (53)$$

then (52) is exactly

$$f_n = p_n * u_{a_n}, \quad \text{up} = p_n * \rho_{a_n}. \quad (54)$$

The repeated-integration approximation therefore does one very specific thing: it keeps the first n dyadic uniform coordinates exact and replaces the remaining self-similar tail $a_n X'$ by a single uniform coordinate $a_n U$.

5.1 The comparison kernel

Definition 5.1 (Uniform-versus-up Peano kernel). For $t \in \mathbb{R}$, define

$$K(t) = \mathbb{E}(U - t)_+ - \mathbb{E}(X - t)_+. \quad (55)$$

The kernel contains every constant needed for the sharp error theorem. We first calculate its basic invariants.

Lemma 5.2 (Second and first absolute moments). *The limiting random variable X satisfies*

$$\mathbb{E}X^2 = \frac{1}{9}, \quad \mathbb{E}|X| = \frac{5}{18}. \quad (56)$$

Consequently,

$$F\left(\frac{1}{4}\right) = \frac{5}{72}. \quad (57)$$

Proof. Independence and centering in (26) give

$$\mathbb{E}X^2 = \frac{1}{3} \sum_{k=1}^{\infty} 4^{-k} = \frac{1}{9}.$$

For fixed $x \in [-1, 1]$ and U uniform on $[-1, 1]$,

$$\mathbb{E}(|x + U| \mid x) = \frac{1}{2} \int_{-1}^1 |x + u| \, du = \frac{1 + x^2}{2}.$$

Using $X \stackrel{d}{=} (X' + U)/2$ gives

$$\mathbb{E}|X| = \frac{1}{2} \mathbb{E}|X' + U| = \frac{1}{4} (1 + \mathbb{E}X^2) = \frac{5}{18}.$$

By symmetry, $\mathbb{E}X_+ = \frac{1}{2} \mathbb{E}|X| = 5/36$. On the other hand, (35) and $F'(y) = 2F(2y)$ imply

$$\mathbb{E}X_+ = \int_0^1 F\left(\frac{1-t}{2}\right) \, dt = 2 \int_0^{1/2} F(y) \, dy = 2F\left(\frac{1}{4}\right),$$

which proves (57). $\square$

Theorem 5.3 (Exact kernel data). *The kernel K is even, continuous, supported in $[-1, 1]$, and nonnegative. For $0 \leq t \leq 1$,*

$$K(t) = \frac{(1-t)^2}{4} - 2F\left(\frac{1-t}{4}\right). \quad (58)$$

Moreover,

$$\boxed{\int_{-1}^1 K(t) \, dt = \frac{1}{9}, \quad \|K\|_{\infty} = K(0) = \frac{1}{9}}, \quad (59)$$

while

$$\|K'\|_{\infty} = \frac{13}{72}. \quad (60)$$

Proof. Evenness follows from symmetry and the fact that U and X have the same mean. Both variables are supported in $[-1, 1]$, so K vanishes outside that interval.

For nonnegativity, let ϕ be convex. From the fixed-point identity $X \stackrel{d}{=} (X' + U)/2$ and convexity,

$$\mathbb{E}\phi(X) \leq \frac{1}{2} \mathbb{E}\phi(X') + \frac{1}{2} \mathbb{E}\phi(U).$$

Since X' has the same law as X , this rearranges to $\mathbb{E}\phi(X) \leq \mathbb{E}\phi(U)$. Thus X is below U in convex order. Applying this to $\phi_t(y) = (y - t)_+$ gives $K(t) \geq 0$.

For $0 \leq t \leq 1$,

$$\mathbb{E}(U - t)_+ = \frac{(1 - t)^2}{4}.$$

Also, by (35),

$$\begin{aligned} \mathbb{E}(X - t)_+ &= \int_t^1 \mathbb{P}(X > s) \, ds = \int_t^1 F\left(\frac{1 - s}{2}\right) \, ds \\ &= 2 \int_0^{(1-t)/2} F(y) \, dy = 2F\left(\frac{1 - t}{4}\right), \end{aligned}$$

which proves (58).

The Peano identity proved in theorem 5.4 below, applied to $h(y) = y^2/2$, gives

$$\int K(t) \, dt = \frac{1}{2}(\mathbb{E}U^2 - \mathbb{E}X^2) = \frac{1}{2}\left(\frac{1}{3} - \frac{1}{9}\right) = \frac{1}{9}.$$

At the origin, theorem 5.2 gives

$$K(0) = \mathbb{E}U_+ - \mathbb{E}X_+ = \frac{1}{4} - \frac{5}{36} = \frac{1}{9}.$$

It remains to see that this is the maximum. Differentiating (58) gives

$$K'(t) = F\left(\frac{1 - t}{2}\right) - \frac{1 - t}{2}. \quad (61)$$

On $[0, 1/2]$, the Fabius function is convex: $F'(y) = 2F(2y)$ is increasing. Since $F(0) = 0$ and $F(1/2) = 1/2$, convexity places its graph below the chord, so $F(y) \leq y$. Therefore $K'(t) \leq 0$ for $t \geq 0$, and $K(0)$ is the maximum.

Finally, put $y = (1 - t)/2$ in (61). The magnitude of K' is the maximum of $y - F(y)$ on $[0, 1/2]$. Its derivative is $1 - 2F(2y)$, which vanishes exactly at $y = 1/4$ because $F(1/2) = 1/2$. Thus

$$\|K'\|_\infty = \frac{1}{4} - F\left(\frac{1}{4}\right) = \frac{1}{4} - \frac{5}{72} = \frac{13}{72}.$$

□

The equality between the mass and the maximum in (59) is the key numerical coincidence behind the clean final rate. The mass controls all sufficiently smooth stages; the maximum controls the borderline stage at which a distributional box-spline derivative becomes an ordinary continuous error function.

5.2 Peano identity and the pointwise error formula

Lemma 5.4 (Peano identity). *Let h' be absolutely continuous on $[-1, 1]$. Then*

$$\mathbb{E}h(U) - \mathbb{E}h(X) = \int_{-1}^1 K(t)h''(t) \, dt. \quad (62)$$

Proof. For $y \in [-1, 1]$,

$$h(y) = h(-1) + h'(-1)(y + 1) + \int_{-1}^1 (y - t)_+ h''(t) \, dt.$$

Take expectations first with $y = U$ and then with $y = X$. The constant terms cancel, and the linear terms cancel because both random variables have mean zero. Fubini's theorem then gives (62). □

Apply the lemma with $h(y) = p_n^{(r)}(x - a_n y)$. Whenever $n \geq r + 3$, the needed weak second derivative is bounded and the result is an ordinary integral.

Proposition 5.5 (Exact pointwise error identity). *Let $r \geq 0$ and $n \geq r + 3$. Then for every $x \in \mathbb{R}$,*

$$f_n^{(r)}(x) - \text{up}^{(r)}(x) = a_n^2 \int_{-1}^1 K(t) p_n^{(r+2)}(x - a_n t) dt. \quad (63)$$

Proof. From (54),

$$f_n^{(r)}(x) = \mathbb{E} p_n^{(r)}(x - a_n U), \quad \text{up}^{(r)}(x) = \mathbb{E} p_n^{(r)}(x - a_n X).$$

Use [theorem 5.4](#); differentiating twice with respect to its argument contributes the factor a_n^2 . $\square$

There is also a distributional form that remains valid at the borderline stage. Define

$$G_a(x) = aK(x/a). \quad (64)$$

Since K'' is the difference between the densities of U and X in the distributional sense,

$$u_a - \rho_a = D^2 G_a. \quad (65)$$

Consequently,

$$f_n - \text{up} = p_n * D^2 G_{a_n}. \quad (66)$$

This identity will handle the first two smoothness thresholds without any appeal to a classical second derivative of p_n .

6 Sharp derivative geometry of the partial densities

The pointwise identity (63) becomes sharp because the derivatives of p_n are disjoint signed translates of a single tail density, each having a flat top of known width and height.

For $m \geq 0$, define the dyadic centers

$$c_{m,j} = -1 + \frac{2j+1}{2^m}, \quad 0 \leq j < 2^m. \quad (67)$$

For $m = 0$, this is the singleton $\{0\}$. At the top piecewise-polynomial derivative order, derivatives below are understood through their bounded almost-everywhere representatives; values at the finitely many knots may be chosen arbitrarily between the adjacent limits and do not affect the L^∞ norm used in this section.

Theorem 6.1 (Derivative decomposition and exact plateaux). *Let $0 \leq m \leq n - 1$. Let $q_{m,n}$ be the density of the tail sum*

$$R_{m,n} = \sum_{k=m+1}^n 2^{-k} U_k. \quad (68)$$

Then, almost everywhere,

$$p_n^{(m)}(x) = 2^{\binom{m}{2}} \sum_{j=0}^{2^m-1} (-1)^{\text{wt}(j)} q_{m,n}(x - c_{m,j}). \quad (69)$$

The translated supports in this sum have disjoint interiors. Moreover,

$$\|p_n^{(m)}\|_\infty = 2^{\binom{m+1}{2}}, \quad (70)$$

and on the central interval of each translated tail,

$$p_n^{(m)}(x) = (-1)^{\text{wt}(j)} 2^{\binom{m+1}{2}} \quad \text{whenever} \quad |x - c_{m,j}| < 2^{-n}. \quad (71)$$

Proof. In the distributional sense,

$$Du_{2^{-k}} = 2^{k-1}(\delta_{-2^{-k}} - \delta_{2^{-k}}). \quad (72)$$

Differentiate the first m factors in $p_n = u_{2^{-1}} * \cdots * u_{2^{-n}}$. The product of the scalar factors is

$$\prod_{k=1}^m 2^{k-1} = 2^{\binom{m}{2}}.$$

The 2^m signed sums of the locations $\pm 2^{-k}$ are precisely the centers $c_{m,j}$, in increasing order, and the sign is $(-1)^{\text{wt}(j)}$. Convolution with the undifferentiated tail gives (69).

Adjacent centers are separated by 2^{1-m} . The support radius of $q_{m,n}$ is

$$\sum_{k=m+1}^n 2^{-k} = 2^{-m} - 2^{-n},$$

so its support diameter is $2^{1-m} - 2^{1-n}$, strictly smaller than the center spacing. Hence the translated interiors are disjoint.

Scaling shows

$$q_{m,n}(x) = 2^m p_{n-m}(2^m x). \quad (73)$$

The density p_s has supremum one: its first factor $u_{1/2}$ has height one, and convolution with a probability density cannot increase the supremum. It actually equals one on $|x| < 2^{-s}$, because the remaining factors have total support radius $1/2 - 2^{-s}$. Thus $q_{m,n}$ has height 2^m and is constant at that height on $|x| < 2^{-n}$. Multiplying by $2^{\binom{m}{2}}$ proves (70) and (71). $\square$

At the missing endpoint $m = n$, the same calculation gives a signed Dirac comb:

$$D^n p_n = 2^{\binom{n}{2}} \sum_{j=0}^{2^n-1} (-1)^{\text{wt}(j)} \delta_{c_{n,j}}. \quad (74)$$

The centers are spaced by $2^{1-n} = 2a_n$.

6.1 A sharp derivative norm for the up-function

The finite plateau geometry also recovers a useful sharp fact about the limit itself.

Corollary 6.2 (Exact derivative peaks of up). *For every $m \geq 0$,*

$$\left\| \text{up}^{(m)} \right\|_{\infty} = 2^{\binom{m+1}{2}}. \quad (75)$$

More precisely,

$$\text{up}^{(m)}(c_{m,j}) = (-1)^{\text{wt}(j)} 2^{\binom{m+1}{2}} \quad (0 \leq j < 2^m). \quad (76)$$

Proof. Take $n \geq m + 1$. Since $\text{up} = p_n * \rho_{a_n}$,

$$\text{up}^{(m)}(x) = \int p_n^{(m)}(x-y) \rho_{a_n}(y) dy.$$

Convolution with a probability density cannot exceed $\left\| p_n^{(m)} \right\|_{\infty}$, giving the upper bound. At $x = c_{m,j}$, the whole support $[-a_n, a_n]$ of ρ_{a_n} lies in the constant plateau (71), so the convolution equals the signed plateau height. This proves both statements. $\square$

Corollary 6.3 (Higher-order flatness at the peak grid). *For every $q \geq 0$, every $0 \leq j < 2^q$, and every $k > q$,*

$$\text{up}^{(k)}(c_{q,j}) = 0. \quad (77)$$

Proof. At $q = 0$, the only point is $c_{0,0} = 0$. On the right of zero, $\text{up}(x) = F(1 - x)$, and the smooth bounded Fabius function is constant on $[1, \infty)$; therefore all positive-order derivatives of up vanish at zero.

Let $q \geq 1$, put $a = 2^{-q}$, and write $\text{up} = p_q * \rho_a$. For $\ell = k - q \geq 1$,

$$\text{up}^{(k)} = (D^q p_q) * \rho_a^{(\ell)}.$$

Insert the signed Dirac comb (74). At $x = c_{q,j}$ every off-diagonal translate is evaluated at distance at least $2a$, outside the support $[-a, a]$ of ρ_a . The diagonal translate is $\rho_a^{(\ell)}(0) = a^{-\ell-1} \text{up}^{(\ell)}(0) = 0$ by the first paragraph. Thus every term vanishes. $\square$

7 The exact convergence rate

We now combine the nonnegative kernel of section 5 with the exact plateaux of section 6.

7.1 All stages after the first admissible one

Theorem 7.1 (Sharp L^∞ error for every derivative). *Let $r \geq 0$ and $n \geq r + 2$. Then*

$$\left\| f_n^{(r)} - \text{up}^{(r)} \right\|_\infty = \frac{4^{-n}}{9} \left\| \text{up}^{(r+2)} \right\|_\infty = \frac{2^{\binom{r+3}{2}}}{9 \cdot 4^n}. \quad (78)$$

At the grid (12), the signed equality (14) holds.

Proof. Put $m = r + 2$.

First suppose $n \geq m + 1$. From (63), the nonnegativity and mass of K , and (70),

$$\begin{aligned} |f_n^{(r)}(x) - \text{up}^{(r)}(x)| &\leq a_n^2 \int K(t) dt \left\| p_n^{(m)} \right\|_\infty \\ &= 4^{-n} \cdot \frac{1}{9} \cdot 2^{\binom{m+1}{2}}. \end{aligned}$$

At $x = c_{m,j}$, every point $c_{m,j} - a_n t$ with $-1 < t < 1$ lies in the constant plateau (71). Hence no inequality remains:

$$f_n^{(r)}(c_{m,j}) - \text{up}^{(r)}(c_{m,j}) = (-1)^{\text{wt}(j)} 4^{-n} 2^{\binom{m+1}{2}} \int K(t) dt.$$

This proves the result for $n \geq m + 1$.

It remains to treat the borderline case $n = m$. Use (66) and move all $m = r + 2$ derivatives to p_m :

$$f_m^{(r)} - \text{up}^{(r)} = (D^m p_m) * G_{a_m}.$$

By (74),

$$f_m^{(r)}(x) - \text{up}^{(r)}(x) = 2^{\binom{m}{2}} a_m \sum_{j=0}^{2^m-1} (-1)^{\text{wt}(j)} K\left(\frac{x - c_{m,j}}{a_m}\right). \quad (79)$$

The copies of K have support radius a_m and their centers are $2a_m$ apart, so their interiors are disjoint. Therefore

$$\left\| f_m^{(r)} - \text{up}^{(r)} \right\|_\infty = 2^{\binom{m}{2}} a_m \|K\|_\infty = 2^{\binom{m}{2}-m} \frac{1}{9}.$$

Since $\binom{m}{2} - m = \binom{m+1}{2} - 2m$, this is exactly the right side of (78) with $n = m$. At $x = c_{m,j}$, use $K(0) = 1/9$ to obtain the signed formula. $\square$

The two cases of the proof use different geometric data:

- for $n \geq r + 3$, the error averages a constant extremal plateau and uses $\int K = 1/9$;
- for $n = r + 2$, the error is a disjoint sum of rescaled kernel copies and uses $\|K\|_\infty = 1/9$.

The equality of those two kernel constants is exactly what makes the same formula valid at the threshold and beyond it.

7.2 The first C^r spline

The first iterate with r continuous derivatives is f_{r+1} . Its error has a different constant, controlled by K' rather than by K .

Theorem 7.2 (Exceptional first stage). *For every $r \geq 0$,*

$$\left\| f_{r+1}^{(r)} - \text{up}^{(r)} \right\|_\infty = 2^{\binom{r+1}{2}} \|K'\|_\infty = \frac{13}{72} 2^{\binom{r+1}{2}}. \quad (80)$$

The supremum is attained at every point $c_{r+2,k}$, and the signed value is (13).

Proof. Set $n = r + 1$. In (66), the total derivative order is $r + 2 = n + 1$. Move n derivatives to p_n and one to G_{a_n} . Since $G'_{a_n}(x) = K'(x/a_n)$, (74) gives

$$f_n^{(r)}(x) - \text{up}^{(r)}(x) = 2^{\binom{n}{2}} \sum_{j=0}^{2^n-1} (-1)^{\text{wt}(j)} K' \left(\frac{x - c_{n,j}}{a_n} \right). \quad (81)$$

Again the supports have disjoint interiors, so the supremum is $2^{\binom{n}{2}} \|K'\|_\infty$. Substitute $n = r + 1$ and (60).

For the signed values, (61) gives $K'(-1/2) = 13/72$ and $K'(1/2) = -13/72$. The two points $c_{n,j} \mp a_n/2$ are respectively $c_{n+1,2j}$ and $c_{n+1,2j+1}$; using $\tau_{n+1}(2j) = \tau_n(j)$ and $\tau_{n+1}(2j+1) = -\tau_n(j)$ in (81) proves (13). $\square$

7.3 Exact C^r convergence

Because the constants in (78) increase strictly with the derivative order, the highest derivative controls the max norm (9).

Corollary 7.3 (Exact C^r norm). *For every $r \geq 0$,*

$$\|f_{r+1} - \text{up}\|_{C^r} = \frac{13}{72} 2^{\binom{r+1}{2}}, \quad (82)$$

and for every $n \geq r + 2$,

$$\boxed{\|f_n - \text{up}\|_{C^r} = \frac{2^{\binom{r+3}{2}}}{9 \cdot 4^n}}. \quad (83)$$

Thus, for every fixed r , the tail of the sequence lies in $C^r(\mathbb{R})$ and converges to up in the C^r norm; the error is exactly geometric from the second admissible stage onward. No finite f_n is itself C^∞ , so this is deliberately stated as eventual convergence in every finite smoothness norm rather than literally as convergence of a sequence inside the Fréchet space $C^\infty(\mathbb{R})$.

Proof. For $n \geq r + 2$, the constants in (78) grow strictly with the derivative order, since the ratio of the order- $(j + 1)$ constant to the order- j constant is 2^{j+3} . Hence the r th derivative controls the

maximum in (9). At the exceptional stage $n = r + 1$, the assertion is immediate for $r = 0$. For $r \geq 1$, the largest lower-order error is the order- $(r - 1)$ error, and its ratio to the order- r error in (80) is

$$\frac{2^{\binom{r+2}{2}} / (9 \cdot 4^{r+1})}{(13/72) 2^{\binom{r+1}{2}}} = \frac{2^{2-r}}{13} < 1.$$

Thus the highest available derivative controls the max norm at the first stage as well. $\square$

For the tent indexing $g_m = f_{m+1}$, this becomes the following direct answer to the piecewise-linear formulation.

Corollary 7.4 (Repeated integration starting from the tent). *Let $g_0(x) = (1 - |x|)_+$ and $g_{m+1} = Tg_m$. For every $r \geq 0$,*

$$\|g_r - \text{up}\|_{C^r} = \frac{13}{72} 2^{\binom{r+1}{2}}, \quad (84)$$

and, for every $m \geq r + 1$,

$$\|g_m - \text{up}\|_{C^r} = \frac{2^{\binom{r+3}{2} - 2}}{9 \cdot 4^m}. \quad (85)$$

In particular, (16) holds.

7.4 The uniform errors and exact extremal values

For $r = 0$, the complete startup behavior is

$$\|f_0 - \text{up}\|_\infty = \frac{1}{2}, \quad \|f_1 - \text{up}\|_\infty = \frac{13}{72}, \quad \|f_n - \text{up}\|_\infty = \frac{8}{9 \cdot 4^n} \quad (n \geq 2). \quad (86)$$

The value for f_0 is immediate from $\text{up}(0) = 1$ and $0 \leq \text{up} \leq 1$. For f_1 , put $y = 1 - |x|$. By (36),

$$f_1(x) - \text{up}(x) = y - F(y), \quad (87)$$

whose absolute maximum is attained at $y = 1/4$ and equals $1/4 - F(1/4) = 13/72$.

The four extremizers from (12) are $\pm 1/4, \pm 3/4$. Since

$$\text{up}(\pm 3/4) = F(1/4) = \frac{5}{72}, \quad \text{up}(\pm 1/4) = 1 - F(1/4) = \frac{67}{72}, \quad (88)$$

we obtain exact finite-stage values for every $n \geq 2$:

$$f_n(\pm 3/4) = \frac{5}{72} + \frac{8}{9 \cdot 4^n}, \quad (89)$$

$$f_n(\pm 1/4) = \frac{67}{72} - \frac{8}{9 \cdot 4^n}. \quad (90)$$

The first few are shown in table 1.

8 The asymptotic error profile

The exact norm formula gives the rate, but the Peano identity also identifies the whole limiting error shape.

Theorem 8.1 (Uniform leading profile). *For every fixed $r \geq 0$,*

$$4^n (f_n^{(r)} - \text{up}^{(r)}) \longrightarrow \frac{1}{9} \text{up}^{(r+2)} \quad \text{uniformly on } \mathbb{R}. \quad (91)$$

Table 1: Sharp uniform errors and two extremal values.

n	$\ f_n - \text{up}\ _\infty$	$f_n(3/4)$	$f_n(1/4)$
0	1/2	–	–
1	13/72	1/4	3/4
2	1/18	1/8	7/8
3	1/72	1/12	11/12
4	1/288	7/96	89/96
5	1/1152	9/128	119/128
6	1/4608	107/1536	1429/1536

Proof. Put $m = r + 2$. From (63),

$$4^n (f_n^{(r)}(x) - \text{up}^{(r)}(x)) = \int K(t) p_n^{(m)}(x - a_n t) dt.$$

For fixed m , $p_n^{(m)} \rightarrow \text{up}^{(m)}$ uniformly. One direct estimate follows from $\text{up} = p_n * \rho_{a_n}$ and the mean-value theorem:

$$\|p_n^{(m)} - \text{up}^{(m)}\|_\infty \leq a_n \mathbb{E}|X| \|p_n^{(m+1)}\|_\infty,$$

for all sufficiently large n . The same derivative bound shows that replacing $p_n^{(m)}(x - a_n t)$ by $p_n^{(m)}(x)$ costs $O(a_n)$ uniformly. Finally use $\int K = 1/9$. $\square$

Combining (91) with (75) predicts the sharp constant in (78). The exact theorem says more: the norm of the leading profile is already the exact norm of the full error for every $n \geq r + 2$.

8.1 A full even-derivative expansion

The Fourier product provides an all-orders refinement. Put

$$\Phi(z) = \prod_{k=1}^{\infty} \text{sinc}(z/2^k), \quad A(z) = \frac{\text{sinc } z}{\Phi(z)}. \quad (92)$$

The quotient is analytic near the origin (indeed on $|z| < 2\pi$) and meromorphic globally. Wherever $\Phi(2^{-n}t) \neq 0$ one has

$$\mathcal{F} f_n(t) = \mathcal{F} \text{up}(t) A(2^{-n}t), \quad (93)$$

and at the exceptional frequencies the product on the right is interpreted by its continuous finite-product extension. Only the Taylor germ at zero is used below. It is convenient to encode that germ by

$$B(w) = A(iw) = \frac{\sinh w/w}{\prod_{k=1}^{\infty} \frac{\sinh(w/2^k)}{w/2^k}} = \sum_{m=0}^{\infty} \beta_m w^{2m}. \quad (94)$$

The first coefficients are

$$\beta_0 = 1, \quad \beta_1 = \frac{1}{9}, \quad \beta_2 = \frac{2}{2025}, \quad \beta_3 = -\frac{1}{2679075}, \quad \beta_4 = \frac{58}{10247461875}. \quad (95)$$

Proposition 8.2 (All-orders C^r expansion). *For fixed integers $r \geq 0$ and $M \geq 1$,*

$$f_n^{(r)} = \sum_{m=0}^{M-1} \beta_m 4^{-mn} \text{up}^{(r+2m)} + O_{r,M}(4^{-Mn}) \quad (96)$$

uniformly on $\mathbb{R}$ as $n \rightarrow \infty$.

Proof. Choose R with $1 < R < 2\pi$ and put $a = 2^{-n}$. On $|z| \leq R$, Taylor's theorem gives

$$A(z) = \sum_{m=0}^{M-1} \beta_m (-1)^m z^{2m} + O_{M,R}(|z|^{2M}). \quad (97)$$

On the low-frequency region $|t| \leq R/a$, multiply this by $(it)^r \mathcal{F} \text{up}(t)$. Since $\mathcal{F} \text{up}$ is a Schwartz function, the L^1 norm of the resulting remainder is at most

$$C_{r,M,R} a^{2M} \int_{\mathbb{R}} |t|^{r+2M} |\mathcal{F} \text{up}(t)| dt = O_{r,M}(a^{2M}).$$

The sign in (97) is exactly the sign in $(it)^{2m} = (-1)^m t^{2m}$, so Fourier inversion produces the displayed derivatives of up .

It remains to control $|t| > R/a$. From (31) and $|\text{sinc } y| \leq \min(1, |y|^{-1})$, for $|t| \geq R2^n$,

$$|\mathcal{F} f_n(t)| \leq 2^{n(n+1)/2+n} |t|^{-(n+1)}. \quad (98)$$

Hence, for fixed r and large n ,

$$\int_{|t| > R2^n} |t|^r |\mathcal{F} f_n(t)| dt \leq C_r R^{r-n} 2^{-n^2/2+(r+3/2)n},$$

which is smaller than a^{2M} for all sufficiently large n . The high-frequency tails of the finitely many $\text{up}^{(r+2m)}$ terms are also $O_{r,M}(a^{2M})$, again because $\mathcal{F} \text{up}$ is Schwartz. Fourier inversion now proves the uniform remainder in (96). $\square$

For explicit coefficient generation, write

$$\log B(w) = \sum_{m=1}^{\infty} \lambda_m w^{2m}. \quad (99)$$

With the standard Bernoulli numbers B_{2m} ,

$$\lambda_m = \frac{2^{2m-1} B_{2m}}{m(2m)!} \frac{4^m - 2}{4^m - 1}. \quad (100)$$

Indeed,

$$\log \frac{\sinh w}{w} = \sum_{m \geq 1} \frac{2^{2m-1} B_{2m}}{m(2m)!} w^{2m}, \quad \sum_{k \geq 1} 4^{-mk} = \frac{1}{4^m - 1};$$

subtracting the logarithms of the denominator factors in (94) gives (100). The coefficients β_m are then obtained recursively from

$$\beta_0 = 1, \quad \beta_m = \frac{1}{m} \sum_{j=1}^m j \lambda_j \beta_{m-j}. \quad (101)$$

Thus the first terms of (96) are

$$f_n^{(r)} = \text{up}^{(r)} + \frac{\text{up}^{(r+2)}}{9 \cdot 4^n} + \frac{2 \text{up}^{(r+4)}}{2025 \cdot 16^n} - \frac{\text{up}^{(r+6)}}{2679075 \cdot 64^n} + O_r(256^{-n}). \quad (102)$$

At the extremal grid (12), all higher derivative terms vanish by [theorem 6.3](#); there the leading term is not merely asymptotic but exactly equal to the full error, as (14) states.

9 Exact computation and numerical use

9.1 A direct exact-arithmetic evaluator

Formula (5) immediately gives an exact rational evaluator at rational arguments. The following compact Python implementation uses arbitrary-precision fractions. It is intended as executable notation rather than as the fastest large- n algorithm.

Listing 1: Exact evaluation of the finite iterates.

```

1 from fractions import Fraction
2 from math import factorial
3
4
5 def tau(n: int, j: int) -> int:
6     if j < 0 or j >= (1 << n):
7         return 0
8     return -1 if (j.bit_count() & 1) else 1
9
10
11 def positive_power(x: Fraction, n: int) -> Fraction:
12     return x**n if x > 0 else Fraction(0)
13
14
15 def f(n: int, x: Fraction) -> Fraction:
16     x = Fraction(x)
17     if n == 0:
18         return Fraction(1, 2) if abs(x) < 1 else Fraction(0)
19
20     coefficient = Fraction(
21         1 << (n * (n + 1) // 2 - 1), factorial(n)
22     )
23     mesh = Fraction(1, 1 << (n - 1))
24
25     return coefficient * sum(
26         (tau(n, j) - tau(n, j - 1))
27         * positive_power(x + 1 - j * mesh, n)
28         for j in range((1 << n) + 1)
29     )

```

Only indices satisfying $j < 2^{n-1}(x + 1)$ contribute, so the loop may be truncated at the active cell. Derivatives are obtained by replacing $n!$ with $(n - r)!$ and the power n with $n - r$, as in (44).

9.2 Conditioning and stable evaluation

The truncated-power formula is exact but can exhibit severe cancellation at large n : individual terms are much larger than their sum. Three alternatives are preferable in floating-point work.

- (i) Use the convolution description (23) and propagate a piecewise-polynomial representation from one level to the next.
- (ii) Use a B-spline/de Boor representation on the dyadic knot sequence rather than the truncated-power basis.
- (iii) In Fourier space, evaluate the finite sinc product (31) and invert it with a quadrature adapted to a compactly supported smooth target.

For certified approximation of up, the sharp error gives an immediate stopping rule. To obtain uniform error at most ε from the tent-indexed sequence, it is necessary and sufficient that

$$m \geq \left\lceil \frac{1}{2} \log_2 \left(\frac{2}{9\varepsilon} \right) \right\rceil \quad (m \geq 1). \quad (103)$$

For a fixed derivative order r , replace the numerator $2/9$ by $2^{\binom{r+3}{2}-2}/9$ and also require $m \geq r + 1$.

10 How this fits the existing ProveIt development

The present approximation is closely related to, but not identical with, two families already developed in the repository.

- The finite convolution and polynomial machinery in `FabiusFunction/EarlyApproximations.lean` provides reusable measure-theoretic and coefficient infrastructure.
- The Thue–Morse spline machinery in `FabiusFunction/FabiusUniformSpline.lean` already contains many of the finite-power sum and block-translation lemmas needed to identify the top derivative pattern.
- The centered finite-CDF spline in the main article approximates the bounded Fabius function. The sequence here approximates the up-density itself and has a different tail replacement and a different sharp rate.

A natural formalization could be organized as follows.

- (1) Define the probability operator T and prove $Th(x) = \int_{2x-1}^{2x+1} h$ for supported integrable h .
- (2) Define X_n recursively and prove the finite random-sum identity (22); derive (23) at the level of measures.
- (3) Introduce the nonuniform box spline and prove the general inclusion–exclusion lemma (37).
- (4) Reuse the finite Thue–Morse product to prove (43), then obtain the exact iterate and derivative formulas.
- (5) Define K by (55). Prove convex domination with the one-line Jensen argument, then formalize its mass, maximum, and derivative maximum.
- (6) Prove the derivative decomposition (69), support separation, and the exact plateaux.
- (7) Combine the Peano identity with the plateau lemma, splitting the proof into the ordinary case $n \geq r + 3$ and the distributional threshold $n = r + 2$.

The analytic core of the sharp theorem is therefore modest. The technically delicate parts for Lean are likely to be bookkeeping for weak derivatives of interval densities and the transfer between measure convolution and classical derivatives. One can avoid most of that friction by proving the borderline formulas directly from finite signed Dirac combs and using ordinary absolutely continuous derivatives only after the threshold.

11 Conclusions

The repeated-integration construction has three mutually reinforcing exact descriptions:

- (i) an averaging recursion $X \mapsto (X + U)/2$;
- (ii) a convolution of dyadically scaled uniform densities;

(iii) a dyadic polynomial spline whose top derivative is a Thue–Morse word.

These descriptions identify the limit as Rvachev's up-function and reduce the finite-stage closed form to a single first-difference Thue–Morse sum. More unexpectedly, the same probabilistic representation yields a completely sharp convergence theorem. The finite stage and the limit differ only in the last scaled tail: a uniform variable replaces an up-distributed variable. Their mass and mean agree, so the error starts at second order in 2^{-n} . The comparison kernel quantifies this replacement exactly, while the partial box spline supplies disjoint extremal plateaux. The outcome is the exact identity

$$\left\| f_n^{(r)} - \text{up}^{(r)} \right\|_{\infty} = \frac{2^{\binom{r+3}{2}}}{9 \cdot 4^n} \quad (n \geq r + 2),$$

not just an upper bound or an asymptotic equivalent.

In the most literal piecewise-linear formulation, starting from the tent and integrating again and again, the answer is especially crisp: after the initial tent, the uniform error is exactly $2/(9 \cdot 4^m)$. Every additional integration divides the worst-case error by four, and the points at which that error is attained are fixed dyadic points rather than moving with the iteration.

References

- [1] kram1032, *Recursive Integration over Piecewise Polynomials: Closed form?*, Mathematics Stack Exchange, question 240687, 19 November 2012, <https://math.stackexchange.com/questions/240687/recursive-integration-over-piecewise-polynomials-closed-form>.
- [2] Nathan Grigg, *Answer to "Recursive Integration over Piecewise Polynomials: Closed form?"*, Mathematics Stack Exchange, 22 November 2012, <https://math.stackexchange.com/a/240805>.
- [3] Juan Arias de Reyna, *An infinitely differentiable function with compact support: Definition and properties*, English translation, arXiv:1702.05442, 2017; original Spanish version in *Revista de la Real Academia de Ciencias Exactas, Físicas y Naturales de Madrid* **76** (1982), 21–38.
- [4] Juan Arias de Reyna, *Arithmetic of the Fabius function*, arXiv:1702.06487; *Integers* **18** (2018), A51.
- [5] Jaap Fabius, *A probabilistic example of a nowhere analytic C^∞ -function*, *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* **5** (1966), 173–174.
- [6] Børge Jessen and Aurel Wintner, *Distribution functions and the Riemann zeta function*, *Transactions of the American Mathematical Society* **38** (1935), 48–88.
- [7] Vladimir Reshetnikov, *ProveIt: Analysis/FabiusFunction*, <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction>, repository snapshot consulted 25 August 2026.